# Verification Plan: 4 bit Alu

**1.DUT Overview**

Module: alu_4bit
Function: 4bit alu with following supported operations and inputs

| Opcode (op) | Operation | Description | Example (A=3, B=2) | Y | carry |
|---|---|---|---|---|---|
| 000 | ADD | Y = A + B | 3 + 2 | 0101 | Carry = 0 |
| 001 | SUB | Y = A - B (two's complement) | 3 - 2 | 0001 | Carry = 0 |
| 010 | AND | Bitwise AND | 3 & 2 | 0010 | 0 |
| 011 | OR | Bitwise OR | 3 \| 2 | 0011 | 0 |
| 100 | XOR | Bitwise XOR | 3 ^ 2 | 0001 | 0 |
| 101 | NOT A | Bitwise negation of A | ~3 | 1100 | 0 |
| 110 | PASS A | Output A | 3 | 0011 | 0 |
| 111 | PASS B | Output B | 2 | 0010 | 0 |

**2.Verification Objectives**

- Check correct outcome for each operation
- Check correct carry bit logic
- Check correct opcode interpretation

**3.Test Strategy**

For each input combination check Y and carry output

**4.Testbench Architecture**

- Simple behavioral reference model to predict expected value
- Compare outputs of reference and test module

**5.Coverage Goals**

- All combinations of input -> 100% coverage

**6.Pass/Fail Criteria**

- All input combination lead to matching outputs between test module and reference model